DEPARTMENT OF THE ARMY
**UNITED STATES MILITARY ACADEMY**
West Point, New York 10996

MADN-MTH 2 November 2026

MEMORANDUM FOR DXDT “Velocity” Scholarship Committee, United States Military Academy, West Point, NY 10996

SUBJECT: Application for AY 2027 DXDT “Velocity” Scholarship

1. Long-term Professional Goals. I intend to serve as an Army officer who turns operational problems into rigorous models and fielded software, and one day to lead the Army Artificial Intelligence Integration Center (AI2C). I plan to commission in the Infantry and seek assignment to the 82nd Airborne Division at Fort Bragg, North Carolina, because the problems worth modeling start at the tactical level. As an Operations Research major with a finance specialization and a minor in Artificial Intelligence, I then plan to join Functional Area 49 as an operations research analyst (49A) or an AI and machine learning specialist (49B) and earn a master’s degree through the Army’s AI Scholars Program at Carnegie Mellon University or the MIT Lincoln Laboratory Fellowship. I also plan to return to teach in the Department of Mathematical Sciences, because last year LTC Kuiper selected me as one of only two plebes to do research with him, and that decision is why I chose Operations Research. As Cadet-in-Charge of MA103 tutoring at the Center for Enhanced Performance, I already lead the tutor staff for a course of 1,010 cadets, and I want to keep doing that work as an instructor.

2. Purpose. This scholarship would let me test my research where it has to work, inside a company that fields decision software and in front of reviewers who can reject it. This past June, on the Technology and Innovation Academic Individual Advanced Development (AIAD) trip through the Department of Social Sciences (SOSH), Palantir’s team in Palo Alto walked us through Project Maven and how it gives commanders at every echelon a common picture to decide from. A few days later at the 4th Infantry Division, a young captain described a fires exercise in which his cell had minutes to choose which targets to strike with too few rounds, and he admitted the hardest part was not the decision itself but getting the right data in front of him in time. At Stanford’s Gordian Knot Center, COL (Ret.) Joseph Felter and Jack Faherty described how hard it is to get new technology into the hands of Army units, and at Meta we met LTC Andrew Bosworth, Meta’s chief technology officer and an Army Reserve officer in Detachment 201.

I recognized the captain’s problem because I had lived a small version of it. When my parents divorced and my family could not afford groceries, I went to work at McDonald’s, often more than 50 hours a week while still in high school, and gave most of each paycheck to my mother for rent and food. I treated every schedule and supply order as a problem worth solving, and that job taught me that a plan is only as good as the data behind it, which is why the captain’s admission stayed with me. Math has never been just numbers to me, and this funding would take

that same kind of work, math that must hold up for real people, to a defense company, a peer-reviewed conference, and a national hackathon where it can one day hold up for Soldiers.

I also learn by building and by failing. With what I kept from those paychecks, I started flipping watches and grew that capital to more than $30,000, then lost all of it in days when I was sold a counterfeit watch. That loss taught me to verify before I trust and to size risk before I take it, lessons that apply as much to a model’s data as to a watch. I went back to building, first with C2 Claude, where 18 cadets and graduate students are working to build software together, and now with my team in the Data4Good competition, where we currently rank first in a 13-state region and are on track to present at Johns Hopkins University. Palantir and Shield AI both grew from startups into companies the Department of Defense relies on, and working inside one would show me how to build, fail, and fix at the speed the Army will need.

3. Near-term Research Goals. My interest in contested logistics began in July 2026 at the Transformation Decision Analysis Center (TDAC) at Fort Lee, where I worked on a reinforcement-learning agent that allocates resupply in Logistics Battle Command, the Army’s sustainment simulation. I had never studied reinforcement learning, so I taught myself from online resources in the weeks before, and once inside the SCIF, with no internet or phone, I worked through the problem with only the code, my notes, and my own reasoning. I found that the agent was scoring well by starving the priority unit, built two diagnostic tools that exposed four reward flaws, and redesigned the reward so the retrained agent more than tripled the share of requests filled and beat a priority-aware baseline across 101 random seeds. That work showed me how a supply system can look successful while failing the unit that needs it most, and it left me with the opposite question, where an adversary would strike a supply network to cut its flow.

That question became my current project with my advisor, LTC Matthew Webb, whose doctoral research at MIT developed a reinforcement-learning method for finding where to cut Ukraine’s rail network. The question at the center of my research is whether an agent trained with his method can find where an adversary would cut a supply network as well as an exact optimization model, so planners can stress-test contested-logistics plans before a conflict. I chose Taiwan’s crude oil supply as my first test case because it serves both the Army’s priorities in the Indo-Pacific and the Dean’s academic theme. Taiwan imports nearly all of its energy, and Iran’s 2026 closure of the Strait of Hormuz showed how quickly one chokepoint can cut supply. The same methods apply to the Army’s own lines of communication, since the Army carries much of the theater sustainment and fuel distribution in a joint fight.

I modeled Taiwan’s crude import routes as a 61-node, 83-edge capacitated network built from open sources, with nine types of adversary action, each with an effort cost and a chance of success, across 59 components an adversary can act on. Each plan is scored by expected max-flow, the share of demand Taiwan still receives after rerouting, which the adversary seeks to minimize. With an effort budget of 100, random plans leave about 90 percent of demand delivered and a greedy plan leaves 70 percent. After 39 hours of training, my agent’s base-case plan matches the exact optimum of 31.7 percent, and the exact model serves as ground truth for every comparison that follows.

My near-term goal this semester is to (1) test whether the agent keeps matching the optimum across six wargaming scenarios, from the base case to lower and higher action costs, higher and lower demand, and a closed Strait of Hormuz, and see how the adversary’s best plan changes, (2) report measured solve times for the agent and the exact model, and (3) report whether a method built for Ukraine’s rail network transfers to a maritime network.

Over the rest of this academic year, I will extend the work to the close-enough traveling salesman problem, in which a route must only pass within reach of each target area rather than through a fixed point. With multiple vessels, it asks how a small fleet of escort or resupply ships could cover every chokepoint the interdiction model flags in the least time, turning the attacker’s plan into a defender’s routes. I will present both results at Projects Day and, if accepted, at the IEEE Systems and Information Engineering Design Symposium (SIEDS).

In my remaining two and a half years at West Point, I will grow this into a research line on contested logistics, with Taiwan as the first case rather than the last. During my Cow year, I will move the model from one commodity to several, such as fuel and munitions sharing the same network, and from known capacities to uncertain ones, then submit the results to the INFORMS Undergraduate Operations Research Prize and the Military Operations Research Society (MORS) Symposium. During my Firstie year, I will extend it across several time periods so a defender can plan resupply week by week, and, with LTC Webb’s help, offer the finished model to the Army’s Contested Logistics Cross-Functional Team. I have built my time at West Point around this work, and when I choose where my evenings and weekends go, I choose the next experiment and the next question. I would rather go deep on one problem that matters than spread myself evenly across everything, and this is the problem I have chosen.

4. Funding Plan and Budget.

a. Funding Plan. I plan to complete one AIAD, attend two conferences, and complete three online courses. AIAD options are listed in priority order, and I use the DXDT program’s planning estimate of $6,700 for whichever is funded. My Stokes Writing Fellowship and Sea to Shining Sea Fellowship arrange trips, conferences, and expos in their own fields but do not fund mathematical research, so this scholarship is the only source for this work. After each item, I list what it returns to (1) me, (2) the Department of Mathematical Sciences (D/Math), and (3) the Army.

   AIAD, Primary. In summer 2027, I will spend $6,700 on housing, food, and transportation for a three-week AIAD with Palantir or Shield AI, the AIADs for which Velocity Scholars receive priority fill. Palantir builds the Maven Smart System that gives commanders a common picture to decide from, and Shield AI builds autonomy software for unmanned aircraft, so either would show me how a mathematical model becomes a fielded decision tool.

   (1) I would learn how an engineering team turns a model into software a commander trusts and acts on, from integrating messy data to designing what the user sees. (2) D/Math would receive a brief to COL Andrew Lee, Director of the Operations Research Program, LTC Webb, and the DXDT committee within 30 days of my return, and a two-

page after-action report that prepares the next Velocity cohort for the same placement. (3) The Army, which bought Maven Smart System and signed a ten-year enterprise agreement with Palantir in 2025, would gain a future officer who knows how that software is built and where its limits are.

AIAD, Alternate. If neither placement is available, I will do a three-week AIAD from 11–31 July 2027 at MIT Lincoln Laboratory in Lexington, MA, a federally funded research and development center (FFRDC) sponsored by the Department of Defense that builds and tests AI and decision-support tools for the military. There I would further my reinforcement-learning skills and apply my reward-design experience to ongoing defense-sponsored projects.

(1) My payoff is learning how professional researchers validate a model before anyone relies on it, through held-out test cases, stress tests, and adversarial checks. (2) I will turn that process into a testing checklist and share it with LTC Webb and faculty advising cadet reinforcement-learning projects. (3) The Army gains an officer trained to catch an AI tool that scores well while failing its users before that tool is fielded.

Conferences. I will attend two conferences in May 2027, presenting my research at one and competing at the other. In early May, I will spend $1,200 on travel, lodging, meals, and author registration to present my Taiwan interdiction paper at SIEDS at the University of Virginia if it is accepted, and I will return those funds if it is not. On 24–25 May, I will spend $1,100 on train travel, two nights of lodging, and meals at federal per diem rates to attend the Special Competitive Studies Project (SCSP) AI+ Expo in Washington, DC, which drew more than 18,000 registered attendees from defense-technology companies and government in 2026. If SCSP runs its National Security Technology Hackathon again, I will enter with a team of cadets.

(1) Peer review teaches me to frame a problem, defend a method, and state its limits, and the hackathon tests whether I can turn mathematics into a working prototype under time pressure. (2) D/Math would gain a peer-reviewed undergraduate publication from LTC Webb’s line of interdiction research, a hackathon team drawn from its Operations Research, Applied Statistics and Data Science (ASDS), and Mathematical Sciences majors, and a list of firms that take cadets for AIADs or sponsor capstone projects. (3) The Army gains open, peer-reviewed analysis of contested logistics, a problem it stood up a Contested Logistics Cross-Functional Team to solve.

Online Courses. During 2027, I will spend $797 on verified certificates for three courses in the MITx Supply Chain Management MicroMasters program, offered by the MIT Center for Transportation & Logistics, Supply Chain Analytics (SC0x, $199), Supply Chain Fundamentals (SC1x, $299), and Supply Chain Design (SC2x, $299). Verified certificates give me graded exams and credit toward the MicroMasters credential, which auditing does not, and I will schedule each course around the summer AIAD.

Supply Chain Analytics is the program’s first course, and it applies probability and regression to supply data, which I will use to fit my model’s chances of success and capacities to shipping data. Supply Chain Fundamentals covers procurement, inventory, transportation, and distribution, helping me model real supply networks more faithfully.

Supply Chain Design covers network design and optimization and teaches how to decide where added capacity or protection best preserves supply under disruption.

(1) Together, these courses move me from identifying supply chain vulnerabilities to building models that defend against them. (2) Their probability and regression content overlaps MA103, so I will bring its practice problems into the tutoring I lead for MA103’s 1,010 cadets. (3) The Army gains a future platoon leader who understands procurement, inventory, and distribution before planning sustainment for Soldiers.

b. Research Travel. AIAD options are listed in priority order. I will complete one AIAD and attend both conferences.

| **Purpose of Trip (AIAD, Conference, other)** | **Destination** | **City, State** | **Dates or # of Days** | **Estimated total Trip costs** |
|---|---|---|---|---|
| AIAD, Primary | Palantir or Shield AI | Per placement | Summer 2027 (21 days) | $6,700 |
| AIAD, Alternate | MIT Lincoln Laboratory | Lexington, MA | 11–31 Jul 2027 | |
| Conference (present paper) | IEEE SIEDS, UVA | Charlottesville, VA | Early May 2027 | $1,200 |
| Conference and hackathon | SCSP AI+ Expo | Washington, DC | 24–25 May 2027 | $1,100 |

c. Equipment, Materials, or Supplies. Three MITx course certificates from MIT (learn.mit.edu), SC0x ($199), SC1x ($299), and SC2x ($299), for a total of $797.

d. Total. $9,797 (one AIAD $6,700, SIEDS $1,200, SCSP AI+ Expo $1,100, online courses $797), within the $10,000 limit.

e. Timeline for Expenditures. From November 2026 to February 2027, I will request my AIAD choices in priority order and submit my paper to SIEDS. I will present at SIEDS in early May 2027, attend the SCSP AI+ Expo on 24–25 May 2027, and complete the AIAD in summer 2027. I will buy each MITx certificate when that course opens, and all funds will be spent by 31 December 2027.

5. The point of contact for this memorandum is CDT Dennis Pezan at (815) 592-6927 or dennis.pezan@westpoint.edu.

Encl
Applicant’s Curriculum Vitae

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